The University of Lahore, CS&IT Department

Theory of Automata

Assignment # 3

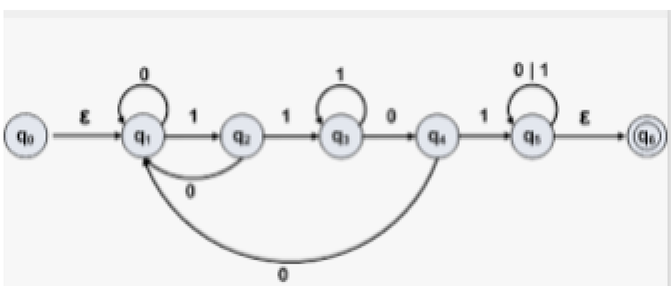
Start Date: 19/12/2024 **Total Marks:** 30

Due Date: 26/12/2024 **Program:** BSCS

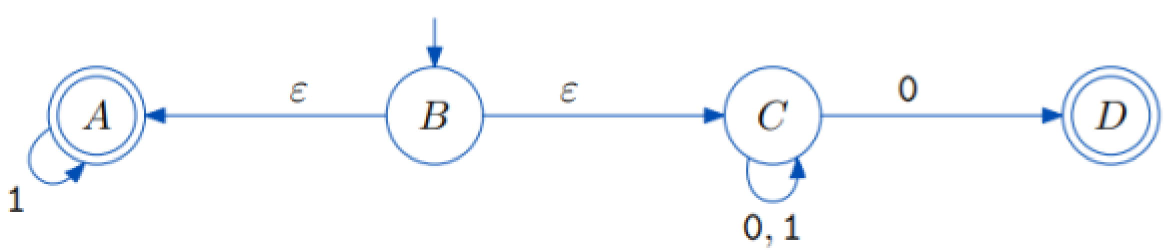
Instructions

1. Understanding of the problems is part of the assignments. So no query please.
2. You will get Zero marks if found any type of cheating.
3. 25 % deduction of over marks on the one day late submission after due date
4. 50 % deduction of over marks on the two day late submission after due date
5. No submission after two days.
6. All questions carry equal marks.

Q1. Suppose q_o is a start state and convert this machine into equivalent Regular expression (All steps are required).



Q2. See the following NFA with null transition and convert it into equivalent DFA (All steps are required).



Q3. Consider the following two DFA's and Construct DFA for the union of Two FA's